30/9/2026

MA5105

10/10

IIT MADRAS

C

Roll No. | I | D | 2 | 5 | S | 0 | 2 | 7 |

PUNEET

Additional Sheet

2.

(a) def ~~sum_product~~ build_list (n):

l = []

always our smallest number will be 111,

and our large number will be nnn,

now we construct a ~~matrix~~ list with simple rule.

$a_0 = 111$, $a_{last} = int(nnn)$

list will be appended starting from 111, then continue to add 1 append, until you reach nnn.

def sum_product (list of nos. n)

sum = 0

for i in range (len (n)):

a = str (i)

for j in range (len (a)):

prod = 1

prod = prod x int (a[j])

sum = sum + prod

return sum

(b) def largest_three_digit(list of nos.)
A

```
    new_list = []
    for i in range (len(A)):
        a = str(i)
        for j in range (len (a)):
            prod = prod * 1
            prod = prod x int (a[j])
        new_list . append (prod)
    new_list . sort()
    return new_list[-1]
```

3

1. So A = QR → upper triangular matrix
   ↓
   orthonormal

(a) $A^{(0)}$

$Q^{(0)}, R^{(0)}$ = qr_decomposition $(A^{(0)})$ → used the supplied func^n

↓

$A^{(1)} = R^{(0)} Q^{(0)}$ [Matrix multiplication]

↓ we now have $A^{(1)}$

$Q^{(1)}, R^{(1)}$ = qr_decomposition $(A^{(1)})$ → used the supplied func^n

↓

$A^{(2)} = R^{(1)} Q^{(1)}$ [Matrix multiplication]

↓ we now have $A^{(2)}$ ready for input

$Q^{(2)}, R^{(2)}$ = qr_decomposition $(A^{(2)})$ → used the supplied func^n

↓

$A^{(3)} = R^{(2)} Q^{(2)}$ [Matrix multiplication]

(b)

~~initialize the A_0~~

```
def qr-decomposition (A):
    '''
    qr iteration method to compute
    the eigen values of a matrix A
    '''
    return Q, R
```

initialize the matrix A as $A^{(0)}$, $k = 0$

~~[max $(a_{ij}^{(k)})$] < tol~~

extract the eigen value of A^0, check $\max(a_{ij}^{(0)})$

while $[\max(a_{ij}^{(k)})] < tol$:

$Q^{(k)}, R^{(k)} = qr\text{-decomposition}(A^{(k)})$

$A^{(k+1)} = R^{(k)} @ Q^{(k)}$

extract the eigen values of $A^{(k+1)}$, and extract the $\max(a_{ij}^{(k)})$ somewhere, because this value has to be checked against the tolerance.

$k = k + 1$ (keep this ready for the next loop)

(7)

once the tolerance will hit and condition becomes false, we come out of the while loop.

now you will already have the eigenvalues for final $A^{(k)}$